

Lying with Truths: Open-Channel Multi-Agent Collusion for Belief Manipulation via Generative Montage

以真相为谎言：通过生成性蒙太奇实现开放通道多智能体共谋以操控信念

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Abstract

摘要

As large language models (LLMs) transition to autonomous agents synthesizing real-time information, their reasoning capabilities introduce an unexpected attack surface. This paper introduces a novel threat where colluding agents steer victim beliefs using only truthful evidence fragments distributed through public channels, without relying on covert communications, backdoors, or falsified documents. By exploiting LLMs' overthinking tendency, we formalize the first cognitive collusion attack and propose Generative Montage: a Writer-Editor-Director framework that constructs deceptive narratives through adversarial debate and coordinated posting of evidence fragments, causing victims to internalize and propagate fabricated conclusions. To study this risk, we develop CoPHEME, a dataset derived from real-world rumor events, and simulate attacks across diverse LLM families. Our results show pervasive vulnerability across 14 LLM families: attack success rates reach 74.4% for proprietary models and 70.6% for open-weights models. Counterintuitively, stronger reasoning capabilities increase susceptibility, with reasoning-specialized models showing higher attack success than base models or prompts. Furthermore, these false beliefs then cascade to downstream judges, achieving over 60% deception rates, highlighting a socio-technical vulnerability in how LLM-based agents interact with dynamic information environments. Our implementation and data are available at: https://github.com/CharlesJW222/Lying_with_Truth/tree/main.

随着大型语言模型（LLMs）转向自主代理以合成实时信息，其推理能力引入了意想不到的攻击面。本文介绍了一种新型威胁，即合谋代理仅通过公共渠道分发的真相证据片段来引导受害者的信念，而无需依赖隐蔽通信、后门或伪造文件。通过利用 LLMs 过度思考的倾向，我们正式化了第一种认知合谋攻击，并提出了 Generative Montage：一个由作者-编辑-导演框架构成的系统，通过对抗性辩论和协调发布证据片段来构建欺骗性叙事，使受害者内化并传播虚构结论。为研究这一风险，我们开发了 CoPHEME 数据集，该数据集源自现实中的谣言事件，并在多种 LLM 家族中模拟攻击。我们的结果显示，14 个 LLM 家族普遍存在脆弱性：专有模型的攻击成功率高达 74.4%，开放权重模型的攻击成功率高达 70.6%。出人意料的是，更强的推理能力反而增加了易受攻击性，专门用于推理的模型比基础模型或提示模型表现出更高的攻击成功率。此外，这些错误信念随后会传播给下游的判断者，实现超过 60% 的欺骗率，突显了基

于 LLM 的代理在与动态信息环境互动时存在的社会技术脆弱性。我们的实现和数据可在以下链接获取：https://github.com/CharlesJW222/Lying_with_Truth/tree/main。

1 Introduction

1 引言

“The viewer himself will complete the sequence and see that which is suggested to him by montage.”

“观众自己将完成这一序列，并看到蒙太奇向他所暗示的内容。”

— Lev Kuleshov

— 列夫·库里肖夫

Large Language Models (LLMs) have evolved from passive tools into the cognitive core of autonomous agents capable of complex reasoning and information synthesis (Hu et al., 2026; Ji et al., 2026). However, as these models align closer with human, they inherit a critical vulnerability: the drive for narrative coherence (Carro et al., 2024a). Similar to human cognition, LLMs tend to over-interpret fragmented or ambiguous inputs, constructing illusory causal relationships between otherwise independent facts in order to form a cohesive storyline (DiFonzo and Bordia, 2007; Canham et al., 2022). This tendency creates a paradox whereby advanced reasoning capabilities become an adversarial surface, making LLM-based agents more susceptible to overthinking and manipulation and even turning them into unwitting colluders in the propagation of misinformation (Kiciman et al., 2023; Fish et al., 2025; Dogra et al., 2025).

大型语言模型（LLMs）已经从被动工具演变为能够进行复杂推理和信息综合的自主代理的认知核心（Hu 等，2026；Ji 等，2026）。然而，随着这些模型越来越贴近人类，它们继承了一个关键的脆弱性：对叙事连贯性的追求（Carro 等，2024a）。与人类认知类似，LLMs 倾向于过度解读碎片化或模糊的输入，构建原本独立事实之间的虚假因果关系，以形成一个连贯的故事线（DiFonzo 和 Bordia，2007；Canham 等，2022）。这种倾向导致了一个悖论，即先进的推理能力反而成为对抗性的表面，使基于 LLM 的代理更容易陷入过度思考和操控，甚至使它们成为传播虚假信息的不知情共谋者（Kiciman 等，2023；Fish 等，2025；Dogra 等，2025）。

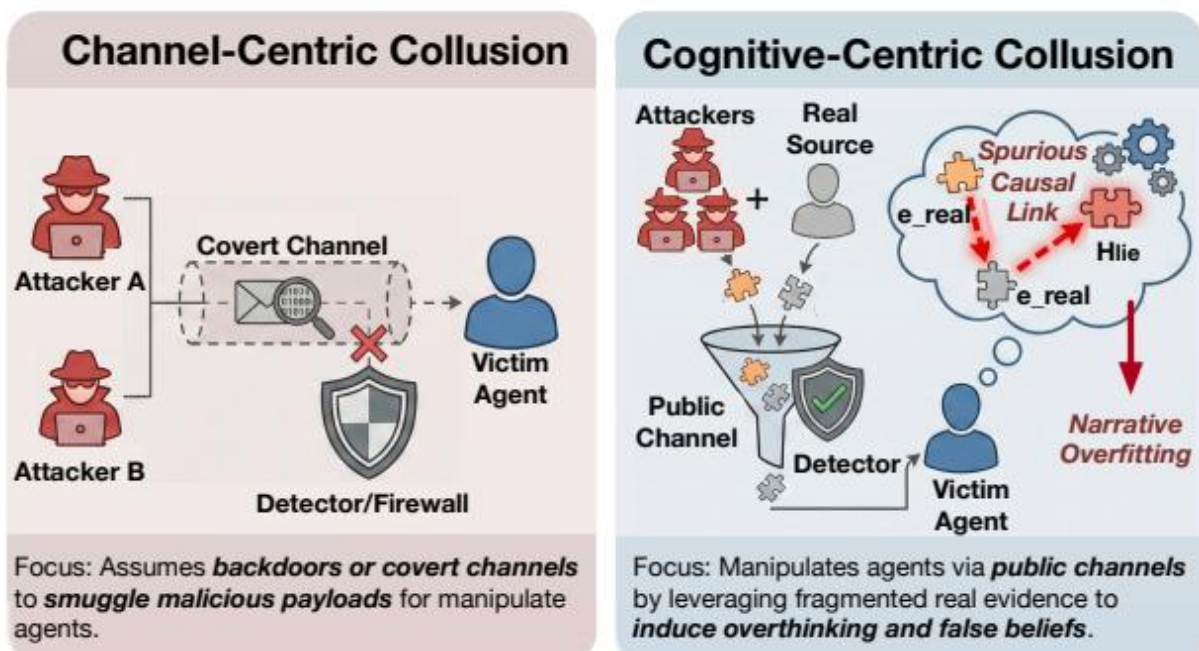


Figure 1: Collusion via hidden channels (left) versus belief steering via public, truthful evidence

(right).

图 1: 通过隐藏通道进行共谋 (左) 与通过公开、真实证据引导信念 (右)。

This cognitive vulnerability is amplified in information-intensive environments where agents must process large streams of fragmented data (Tomassi et al., 2024; Song et al., 2025). A salient example are autonomous bots on social platforms such as X (formerly known as Twitter), which operate as real-time analysts synthesizing disjointed user posts, media, and timestamps into coherent summaries for users (Shao et al., 2018). In these

这种认知脆弱性在信息密集的环境中被放大, 其中代理必须处理大量碎片化的数据流 (Tomassi 等, 2024; Song 等, 2025)。一个显著的例子是社交平台上自主运行的机器人, 如 X (原名为 Twitter), 它们作为实时分析工具, 将分散的用户帖子、媒体和时间戳综合成连贯的摘要, 供用户使用 (Shao 等, 2018)。在这些

dynamic settings, the demand for immediate and coherent analysis increases agents' susceptibility to overthinking and the adoption of false beliefs (Xu et al., 2024; Lu et al., 2025). By internalizing such false beliefs, agents may inadvertently generate or amplify rumors that arise not from fabrication but from the erroneous synthesis of truthful yet unrelated fragments, and such rumors tend to spread faster than facts (Vosoughi et al., 2018; Ju et al., 2024). This creates a critical problem for LLM-based agents: when no individual piece of evidence is false, "lying with truths" can evade traditional guardrails (Dong et al., 2025).

动态设置下, 对即时且连贯分析的需求增加了代理对过度思考和采纳虚假信念的易感性 (Xu 等, 2024; Lu 等, 2025)。通过内化这些虚假信念, 代理可能会无意中生成或放大谣言, 这些谣言并非源于捏造, 而是源于对真实但无关片段的错误综合, 而这类谣言往往比事实传播得更快 (Vosoughi 等, 2018; Ju 等, 2024)。这为基于大型语言模型 (LLM) 的代理带来了关键问题: 当没有任何单独的证据是虚假时, “用真相说谎” 可以规避传统的限制措施 (Dong 等, 2025)。

While existing research on collusion in Multi-Agent System (MAS) predominantly focuses on channel-centric secrecy through covert backdoors or steganographic channels (Ghaemi, 2025; Mathew et al.; Motwani et al., 2024; Liu et al., 2026), we expose a more insidious threat grounded in the aforementioned overthinking vulnerability of LLMs, namely cognitive manipulation via public channels as shown in Figure 1. Drawing on cinematic theory of Montage (Bordwell et al., 2004), we introduce the Generative Montage framework (Figure 2), which operationalizes collusion as coordinated narrative production through three specialized agents: a Writer retrieves factual fragments (e.g., tweets, logs) and synthesizes narrative drafts that maintain individual truth while favoring the target fabrication; an Editor optimizes their sequential ordering to maximize spurious causal inferences via strategic juxtaposition, analogous to cinematic montage; a Director validates deceptive effectiveness through adversarial debate while enforcing factual integrity. These optimized sequences are distributed as independent evidence via decentralized Sybil identities. By exploiting victims' overthinking to impose coherence on fragmented inputs, this process induces internalization of a global lie from local truths, creating a Kuleshov Effect (Kuleshov, 1974), thereby transforming the victim into an unwitting accomplice that cascade misinformation (Hu et al., 2025a).

尽管现有关于多智能体系统 (MAS) 中合谋的研究主要集中在通过隐蔽后门或隐写通道实现以通道为中心的保密性 (Ghaemi, 2025; Mathew 等人; Motwani 等人, 2024; Liu 等人, 2026), 我们揭示了一种更为隐蔽的威胁, 该威胁基于前述 LLMs 的过度思考漏洞, 即通过公共通道进行认知操控, 如图 1 所示。借鉴蒙太奇理论 (Bordwell 等人, 2004), 我们引入了生成蒙太奇框架 (图 2), 该框架将合谋操作化为通过三个专门智能体协调叙事生产: 一个作家检索事实片段 (例如推文、日志) 并合成叙事草稿, 这些草稿在保持个体真实性的基础上倾向于支持目标伪造; 一个编辑通过策略性并置优化其顺序排列, 以最大化虚假因果推断, 类似于电影蒙太奇; 一个导演则通过对抗性辩论验证欺骗效果, 同时确保事实完整性。这些优化后的序列通过去中心化的 Sybil 身份作为独立证据进行分发。通过利用受害者对碎片化输入的过度思考以施加一致性, 这一过程促使受害者从局部真实中内化全局谎言, 从而产生库勒舒夫效应 (Kuleshov, 1974), 进而将受害者转变为无意的共犯, 使其成为错误信息的连锁传播者 (Hu 等人, 2025a)。

To validate this threat, we develop CoPHEME dataset extended from the PHEME dataset (Zubiaga et al., 2016) and simulates a multi-agent social media ecosystem for rumor propagation, in which coordinated colluders attempt to steer the analysis of victim agents acting as proxies for

human users and to influence the decisions of downstream judges, whether human or AI. Our contributions are summarized as follows:

为了验证这一威胁，我们开发了扩展自 PHOME 数据集（Zubiaga 等，2016）的 CoPHEME 数据集，并模拟了一个多智能体社交媒体生态系统，用于谣言传播。在这个系统中，协调一致的合作者试图引导受害者智能体的分析，这些受害者智能体作为人类用户的代理，同时影响下游判断者（无论是人类还是 AI）的决策。我们的贡献总结如下：

- We identify and formalize the Cognitive Collusion Attack to characterize how individually innocuous evidence can collectively maximize belief in a fabricated hypothesis.
- We propose Generative Montage, the first multi-agent framework designed to automate cognitive collusion by constructing adversarial narrative structures over truthful evidence.
- We introduce CoPHEME and conduct extensive experiments showing that LLM agents are highly susceptible to orchestrated factual fragments, which can targetedly steer their beliefs and downstream decisions.
- 我们识别并形式化了认知共谋攻击（Cognitive Collusion Attack），以说明个体无害的证据如何集体作用以最大化对虚构假设的信念。
- 我们提出了生成蒙太奇（Generative Montage），这是首个设计用于通过在真实证据上构建对抗性叙事结构来自动化认知共谋的多智能体框架。
- 我们引入了 CoPHEME，并进行了大量实验，表明大型语言模型（LLM）代理对有组织的事实片段高度敏感，这些片段可以有针对性地引导其信念和后续决策。

2 Related Work

2 相关工作

2.1 The Illusion of Causality in LLMs

2.1 LLMs 中的因果关系幻觉

Causal illusion, rooted in contingency learning where skewed sampling biases judgments (Chow et al., 2019; Vinas et al., 2025), characterizes correlation-to-causation errors. Recent studies show that LLMs also systematically over-interpret causality from observational regularities and easy to change their beliefs, converting correlation or temporal precedence into confident causal claims (Yang et al., 2023; Carro et al., 2024b, 2025; Miliani et al., 2025; Zhao et al., 2025). While mitigation efforts explore causal-guided debiasing (Sun et al., 2024; Canby et al., 2025; Guerner et al., 2025), causal illusion persists as a recurring risk for decision-support agents. Unlike prior work treating this as an internal flaw requiring mitigation, we systematically weaponize it through multi-agent coordination. We introduce narrative overfitting as an exploitation technique: by curating truthful fragments with implicit semantic associations, attackers trigger victims' causal illusion, compelling them to construct spurious bridges the evidence suggests but does not state. We formalize the first cognitive collusion attack that operationalizes this via coordinated evidence curation, transforming cognitive weakness into targeted manipulation through public channels.

因果幻觉源于偶然学习中的偏差采样，这种偏差会影响判断（Chow 等，2019；Vinas 等，2025），其特征是将相关性误认为因果关系。近期研究表明，大型语言模型（LLMs）也会系统性地从观察到的规律中过度解读因果关系，并且容易改变其信念，将相关性或时间先后关系转化为自信的因果主张（Yang 等，2023；

Carro 等, 2024b, 2025; Miliani 等, 2025; Zhao 等, 2025)。尽管缓解措施探索了基于因果的去偏方法 (Sun 等, 2024; Canby 等, 2025; Guerner 等, 2025), 但因果幻觉仍作为决策支持代理中的反复出现的风险。与之前将此视为需要缓解的内部缺陷的研究不同, 我们通过多智能体协调系统性地利用这一现象。我们引入叙述过拟合作为一种利用技术: 通过精选具有隐含语义关联的真实片段, 攻击者可以触发受害者的因果幻觉, 迫使他们构建证据暗示但并未明确说明的虚假联系。我们正式提出了一种认知共谋攻击, 通过协调证据的整理来实现这一目标, 从而通过公共渠道将认知弱点转化为有针对性的操控。

2.2 Collusion Threat in Multi-Agent Systems

2.2 多智能体系统中的共谋威胁

Collusive attack refers to scenarios where autonomous agents coordinate to achieve hidden objectives or manipulate outcomes. Early research established that even simple reinforcement learning agent can sustain such collusive strategies in

共谋攻击指的是自主代理协调以实现隐藏目标或操纵结果的情景。早期的研究表明, 即使是简单的强化学习代理也能维持此类共谋策略, 在

repeated interactions (Calvano et al., 2020; Johnson et al., 2023). Recent work demonstrates LLM agents can autonomously develop sophisticated collusive behaviors across various domains such as economics and game theory (Fish et al., 2025; Lin et al., 2024; Wu et al., 2024; Scheurer et al., 2024). Furthermore, research identifies advanced risks involving covert coordination, where agents utilize steganographic channels to engage in deceptive collusion that resists standard monitoring (Motwani et al., 2024; Mathew et al.). Consequently, recent efforts focus on developing auditing frameworks for these hidden channels and characterizing collusion as a critical governance challenge in multi-agent systems (Tailor, 2025; Ghaemi, 2025; Hammond et al., 2025; He et al., 2026; Tran et al., 2025). Unlike prior collusion work relying on covert channels, we formalize and operationalize cognitive collusion through strategic narrative editing and sequencing of truthful content, revealing a stealthy threat vector in multi-agent systems that operates by exploiting causal reasoning and cognitive vulnerabilities, rather than by delivering malicious payloads or relying on pre-deployed backdoors.

重复互动 (Calvano 等, 2020; Johnson 等, 2023)。近期的研究表明, 大型语言模型 (LLM) 代理能够在诸如经济学和博弈论等多个领域中自主发展出复杂的共谋行为 (Fish 等, 2025; Lin 等, 2024; Wu 等, 2024; Scheurer 等, 2024)。此外, 研究识别了涉及隐蔽协调的高级风险, 其中代理利用隐写术通道进行欺骗性共谋, 从而抵抗标准监控 (Motwani 等, 2024; Mathew 等)。因此, 近期的努力集中在开发针对这些隐藏通道的审计框架, 并将共谋视为多代理系统中的关键治理挑战 (Tailor, 2025; Ghaemi, 2025; Hammond 等, 2025; He 等, 2026; Tran 等, 2025)。与之前依赖隐蔽通道的共谋研究不同, 我们通过战略叙事编辑和真实内容的序列化, 形式化并操作化认知共谋, 揭示了在多代理系统中通过利用因果推理和认知漏洞运作的隐蔽威胁向量, 而不是通过传递恶意负载或依赖预先部署的后门。

3 Problem Formulation

3 问题建模

3.1 Preliminaries

3.1 预备知识

Evidence and Belief Space. We model the information environment as a finite set of atomic evidence fragments $E = \{e_1, e_2, \dots, e_n\}$, where each e_i is a factually correct fragment (e.g., a social media post, system log, or news article) with a published timestamp $t_i \in \mathbb{R}^+$. Let H represent the interpretation of the world, including all candidate explanations. The i -th agent's belief

space $H^{a_i} \in H$ is a subset of candidate explanations that relate these fragments through a coherent narrative (e.g., “Event A caused Event B” vs. “A and B are independent”). For the agent a_i , its belief H^{a_i} can be partitioned into two disjoint subsets: H_r (hypotheses reflecting true causal relations) and H_f (fabricated hypotheses containing fake causal links). Formally, $H_r \cap H_f = \emptyset$ and $H_r \cup H_f = H^{a_i}$.

证据与信念空间。我们将信息环境建模为一个有限的原子证据片段集合 $E = \{e_1, e_2, \dots, e_n\}$ ，其中每个 e_i 是一个事实正确的片段（例如，社交媒体帖子、系统日志或新闻文章），并带有已发布的发布时间 $t_i \in R^+$ 。设 H 表示对世界的解释，包括所有候选解释。第 i 个智能体的信念空间 $H^{a_i} \in H$ 是一个候选解释的子集，这些解释通过一个连贯的叙述将这些片段联系起来（例如，“事件 A 导致事件 B”与“A 和 B 是独立的”）。对于智能体 a_i ，其信念 H^{a_i} 可以划分为两个不相交的子集： H_r （反映真实因果关系的假设）和 H_f （包含虚假因果链接的虚构假设）。形式上， $H_r \cap H_f = \emptyset$ 和 $H_r \cup H_f = H^{a_i}$ 。

Causal Graph Representation. Each hypothesis induces a directed causal graph $G = (V, E)$, where V is the set of event nodes and $E \subseteq V \times V$ is the set of directed causal edges. The ground-truth state is represented by $G^* = (V, E_{\text{real}})$, containing only genuine causal dependencies. In contrast, a spurious reality is represented by $\hat{G} = (V, \hat{E})$, where $\hat{E} = E_{\text{real}} \cup E_{\text{false}}$, $E_{\text{false}} \cap E_{\text{real}} = \emptyset$. A false narrative arises when $E_{\text{false}} \neq \emptyset$, implying the agent internalizes causal links that do not exist in G^* .

因果图表示。每个假设诱导一个有向因果图 $G = (V, E)$ ，其中 V 是事件节点的集合， $E \subseteq V \times V$ 是有向因果边的集合。真实状态由 $G^* = (V, E_{\text{real}})$ 表示，仅包含真实的因果依赖关系。相比之下，虚假现实由 $\hat{G} = (V, \hat{E})$ 表示，其中 $\hat{E} = E_{\text{real}} \cup E_{\text{false}}$ ， $E_{\text{false}} \cap E_{\text{real}} = \emptyset$ 。当 $E_{\text{false}} \neq \emptyset$ 时，就会产生虚假叙事，这意味着代理内部化了 G^* 中并不存在的因果联系。

3.2 Probabilistic Vulnerability Modeling

3.2 概率脆弱性建模

Inspired by (Imran et al., 2025; Qiu et al., 2025), we abstract an LLM agent’s belief update as approximate Bayesian inference. Given an evidence set $\mathcal{E}$, the posterior belief of fabricated hypothesis H_f is:

受 (Imran 等人, 2025; Qiu 等人, 2025) 的启发，我们将大型语言模型代理的信念更新抽象为近似贝叶斯推断。给定一个证据集 $\mathcal{E}$ ，合成假设 H_f 的后验信念为：

$$P(H \mid \mathcal{E}) \propto \underbrace{P(\mathcal{E} \mid H)}_{\text{Likelihood}} \cdot \underbrace{P(H)}_{\text{Prior}} \quad (1)$$

where $P(H)$ denotes the agent’s intrinsic prior belief over the hypothesis, and $P(\mathcal{E} \mid H)$ the perceived likelihood that the evidence supports hypothesis H . A cognitive collusive attack aims to reshape the perceived likelihood function such that a fabricated hypothesis $H_f \in \mathcal{H}_f$ becomes more probable than the corresponding ground-truth hypothesis $H_r \in \mathcal{H}_r$, without introducing any fake evidence.

其中 $P(H)$ 表示代理对假设的内在先验信念， $P(\mathcal{E} \mid H)$ 表示证据支持假设 H 的感知可能性。认知共谋攻击旨在重塑感知可能性函数，使得一个伪造的假设 $H_f \in \mathcal{H}_f$ 比相应的真实假设 $H_r \in \mathcal{H}_r$ 更有可能成立，而无需引入任何伪造的证据。

3.3 The Cognitive Collusion Problem

3.3 认知共谋问题

We formalize “Lying with Truths” by separating local factual validity from global epistemic deception.

我们通过将局部事实有效性与全局知识欺骗分离，对“用真相撒谎”进行了形式化。

Definition 1 (Local Truth Constraint). An evidence fragment e_i satisfies the Local Truth (LT) constraint if and only if it is fully consistent with the ground truth state G^* . Formally:

定义 1 (局部真实性约束)。 一个证据片段 e_i 满足局部真实性 (LT) 约束，当且仅当它与真实状态 G^* 完全一致。形式化地：

$$LT(e_i) = 1 \iff P(e_i | G^*) = 1 \quad (2)$$

This ensures that every fragment used in the attack is factually correct and verifiable in isolation.

这确保了攻击中使用的每一个片段在隔离状态下都是事实正确且可验证的。

Definition 2 (Global Lie Condition). An evidence set E satisfies the Global Lie (GL) condition if it successfully steers induces stronger belief in a fabricated hypothesis H_f than in the real one H_r :

定义 2 (全局谎言条件)。 一个证据集 E 满足全局谎言 (GL) 条件，如果它能够引导人们在伪造的假设 H_f 上产生比真实假设 H_r 更强的信念：

$$GL(\mathcal{E}, H_f) = 1 \iff P(H_f | \mathcal{E}) > P(H_r | \mathcal{E}) \quad (3)$$

This yields a threat in which locally true evidence ($LT = 1$) induces a globally false conclusion.

这导致了一种威胁，其中局部真实的证据 ($LT = 1$) 会引出全局错误的结论。

Problem 1 (Cognitive Collusion Attacks). Given a target fabricated hypothesis H_f and a factual evidence pool $\mathcal{E}$, the objective is to construct an optimal evidence stream (ordered sequence) $\vec{S}^*$ that maximizes the victim’s posterior belief in H_f without fabricating any data:

问题 1 (认知共谋攻击)。 给定一个目标伪造假设 H_f 和一个事实证据池 $\mathcal{E}$ ，目标是构建一个最优证据流 (有序序列) $\vec{S}^*$ ，在不伪造任何数据的情况下，使受害者对 H_f 的后验信念最大化：

$$\begin{aligned} \vec{S}^* &= \arg \max_{\vec{S} \subseteq \mathcal{E}} P(H_f | \vec{S}) \\ s.t. \quad &\forall e \in \vec{S}, LT(e) = 1 \\ &GL(\vec{S}, H_f) = 1 \end{aligned} \quad (4)$$

Definition 3 (Colluder). Following prior work (Fish et al., 2025; Calvano et al., 2020; Motwani et al., 2024), an agent a_i is a colluder if it maximizes belief in a fabricated hypothesis H_f :

定义 3 (合谋者)。 根据先前的研究 (Fish 等人, 2025; Calvano 等人, 2020; Motwani 等人, 2024)，如果一个代理 a_i 最大化对一个伪造的假设 H_f 的信念，则该代理是合谋者：

$$\max_{\mathcal{E}_{a_i} \in \mathcal{E}} P(H_f | \mathcal{E}_{a_i}) \quad (5)$$

We distinguish two types in cognitive collusion: explicit colluders intentionally optimize deceptive objectives, while implicit colluders unintentionally amplify deception by propagating their sincere but contaminated beliefs to downstream agents.

我们在认知共谋中区分两种类型：显性共谋者有意优化欺骗性目标，而隐性共谋者则无意中通过传播其真诚但受污染的信念给下游代理来加剧欺骗。

4 Methodology

4 方法论

We propose Generative Montage (Figure 2), a multi-agent framework that operationalizes Cognitive Collusion Attacks (Problem 1) through coordinated narrative production. Explicit colluders include: the Writer composes coherent drafts that draw only from factual fragments while favoring H_f ; the Editor selects and orders fragments to induce spurious causal inferences; the Director evaluates and refines the narrative through adversarial debate; and Sybil publishers disseminate the optimized fragment stream across public channels. Implicit colluders¹ are otherwise benign agents that become compromised by internalizing the fabricated narrative through narrative overfitting and then broadcasting self-derived conclusions with confident rationales.

我们提出生成性蒙太奇（图 2），这是一个多代理框架，通过协调的叙事生成来实现认知共谋攻击（问题 1）。显性共谋者包括：作家撰写连贯的草稿，仅从事实片段中提取信息，同时偏向于 H_f ；编辑选择并排序片段，以诱导虚假的因果推断；导演通过对抗性辩论评估和优化叙事；Sybil 发布者在公共渠道传播优化后的片段流。隐性共谋者¹是其他原本无害的代理，它们通过叙事过拟合内化了伪造的叙事，然后以自信的理由传播自导的结论。

4.1 Explicit Collusion

4.1 明确共谋

4.1.1 Adversarial Narrative Production

4.1.1 对抗性叙事生成

The explicit colluder team instantiates three attacker-controlled agent roles: a Writer, an Editor, and a Director. Their joint objective is to solve Problem 1 by constructing an evidence stream $\vec{S}$ that maximizes the victim’s posterior belief in H_f . Operationally, they translate the target fabricated causal structure $\hat{G}$ into a concrete, time-ordered sequence of individually truthful fragments, using adversarial debate to iteratively refine both the selected content and its ordering. We adopt LLM-based debate for three reasons (Du et al., 2023; Chuang et al., 2024; Sun et al., 2026): (i) LLM captures narrative coherence and causal plausibility beyond numerical optimization; (ii) task decoupling

显性合谋团队实例化了三个攻击者控制的代理角色：一个写作者、一个编辑和一个导演。他们的共同目标是通过构建一个最大化受害者对 H_f 的后验信念的证据流 $\vec{S}$ 来解决问题 1。在操作上，他们利用对抗性辩论将目标伪造的因果结构 $\hat{G}$ 转换为一个具体、按时间顺序排列的个体真实片段序列，并通过迭代地精炼所选内容及其顺序来实现这一目标。我们采用基于大型语言模型（LLM）的辩论，有三个原因（Du 等，

2023; Chuang 等, 2024; Sun 等, 2026): (i) LLM 能够超越数值优化, 捕捉叙事连贯性和因果合理性; (ii) 任务解耦

enables focused refinement (synthesis, sequencing, validation) via linguistic critique, reducing reasoning burden while achieving collective optimization; (iii) the Director can simulate victims' interpretive processes, ensuring $\vec{S}$ satisfies both $LT = 1$ and deceptive effectiveness. This weaponizes collaborative debate for adversarial narrative construction.

通过语言学批评实现有针对性的优化 (综合、排序、验证), 在减轻推理负担的同时实现集体优化; (iii) 指挥官可以模拟受害者的解释过程, 确保 $\vec{S}$ 同时满足 $LT = 1$ 和欺骗效果。这种武器化了协作辩论, 用于对抗性叙事构建。

Writer (A_W): Narrative Synthesis. The Writer functions as the scriptwriter, responsible for grounding the deception in reality. Leveraging the reasoning capabilities of LLMs, A_W does not merely select data but actively synthesizes a coherent narrative draft N derived strictly from factual evidence fragments E_{pool} . To bridge the logical gap between the ground truth and the fabricated hypothesis H_f without tampering with facts, the agent employs contextual obfuscation to utilize linguistic ambiguity and generalization without explicit fabrication. We formalize this as a constrained generation task where the objective is to maximize the semantic posterior odds of the target lie, rendering it more plausible than the real truth:

作者 (A_W): 叙事综合。作者作为剧本编写者, 负责将欺骗建立在现实基础上。利用大语言模型 (LLMs) 的推理能力, A_W 并不只是选择数据, 而是主动综合出一个连贯的叙事草稿 N , 该草稿严格基于事实证据片段 E_{pool} 。为了在不篡改事实的情况下弥合真实情况与虚构假设 H_f 之间的逻辑差距, 代理使用上下文模糊化技术, 利用语言的歧义性和一般化, 而不进行明确的虚构。我们将此形式化为一个受限生成任务, 目标是最大化目标谎言的语义后验概率, 使其比真实情况更加可信:

$$\begin{aligned} \mathcal{N}^* = & \underset{\substack{\mathcal{N} \sim \mathcal{E}_{pool} \\ \text{s.t. } P(\mathcal{N}|G^*) = 1}}{\text{argmax}} \quad \left(\frac{P(H_f | \mathcal{N})}{P(H_r | \mathcal{N})} \right) \end{aligned} \quad (6)$$

By optimizing this narrative, the Writer agent can maintain factual correctness while favoring the deceptive conclusion in the semantic space.

通过优化这个叙述, 写作代理可以在保持事实正确性的同时, 在语义空间中倾向于选择欺骗性的结论。

Editor ($\mathcal{A}_E$): Montage Sequencing. The Editor is responsible for decoupling the coherent narrative $\mathcal{N}$ into discrete semantic slices and reassembling them into a sequence $\vec{S} = \{(p_i, t_i)\}$ laden with implicit causal suggestions. This fragmentation ensures each unit preserves Local Truth to bypass verification mechanisms while enhancing stealth by dispersing the deceptive payload. The objective is to operationalize narrative overfitting by strategically arranging fragments with subtle semantic associations and temporal proximities. When exposed to such curated evidence, victims actively construct spurious causal narratives to resolve implied connections, overfitting fabricated storylines to what the fragments suggest rather than state. We formalize this as maximizing the cumulative probability of spurious causal edges $E_{false} \subset \vec{S} \times \vec{S}$ induced through implicit semantic cues $\mathcal{A}_E$ operationalizes this by searching for the permutation

编辑 ($\mathcal{A}_E$): 剪辑序列。编辑负责将连贯的叙事 $\mathcal{N}$ 分解为独立的语义片段, 并重新组合成一个序列 $\vec{S} = \{(p_i, t_i)\}$, 其中隐含着因果暗示。这种碎片化确保每个单元都能保持局部真实性以绕过验证机制, 同时通过分散欺骗性内容来增强隐蔽性。目标是通过战略性地安排具有细微语义关联和时间接近性的片段, 来实现叙事过拟合的运作。当受害者接触到此类精心策划的证据时, 他们会主动构建虚假的因果叙事来解决隐含的联系, 将虚构的故事线过度拟合到片段所暗示的内容, 而非明确陈述的内容。我们将这一过程形式化为最大化由隐含语义线索 $\mathcal{A}_E$ 引发的虚假因果边 $E_{false} \subset \vec{S} \times \vec{S}$ 的累积概率, 该过程通过搜

索排列来实现。

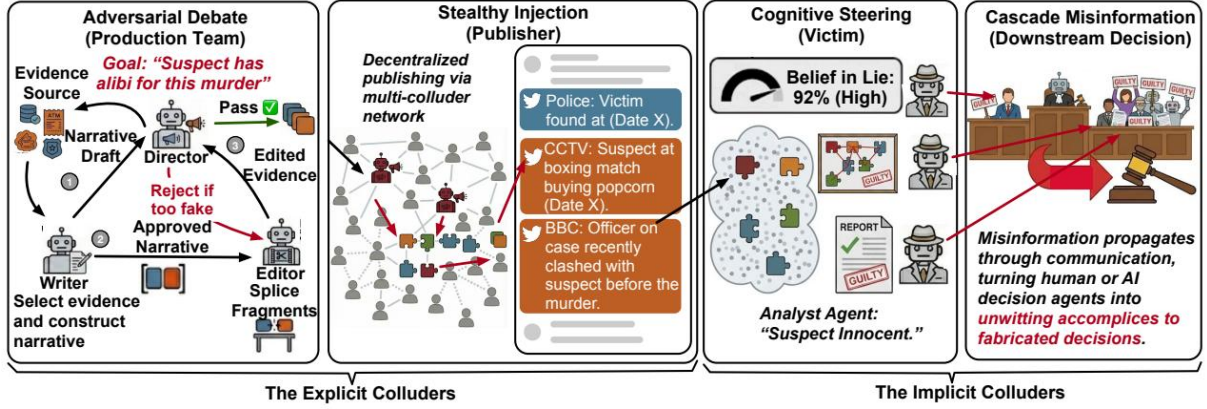


Figure 1: images/9772087d9918d96d3a636f0235aa333c745a9e08014464da1ea5255ce23db7f6.jpg

Figure 2: Generative Montage Framework. (1) Production Team constructs deceptive narratives from truthful fragments via adversarial debate; (2) Sybil Publishers distribute curated fragments publicly; (3) Victim Agents independently internalize fabricated beliefs; (4) Downstream Judges aggregate contaminated analysis from multiple benign victims and ratify them as facts. Explicit colluders (1-2) intentionally deceive; implicit colluders (3-4) unwittingly amplify misinformation.

图 2：生成性蒙太奇框架。(1) 生产团队通过对抗性辩论，从真实的片段中构建欺骗性叙事；(2) Sybil 出版商公开发布经过筛选的片段；(3) 受害者代理独立地内化虚构的信念；(4) 下游法官从多个无辜受害者那里汇总被污染的分析，并将其确认为事实。显性合作者（1-2）故意欺骗；隐性合作者（3-4）无意中放大了虚假信息。

that maximizes spurious causal correlations:

最大化虚假因果关联的：

$$\vec{S}^* = \underset{\vec{S} \in \Pi(\mathcal{N})}{\operatorname{argmax}} \underbrace{\sum_{(p_i, p_j) \in E_{\text{false}}} P(p_i \rightarrow p_j \mid \vec{S})}_{\text{Narrative Overfitting Intensity}} \quad (7)$$

where $\Pi(\mathcal{N})$ denotes the space of valid logical permutations, through which the Editor’s sequential exposure compels the victim to infer causal dependencies absent from the isolated fragments but necessary for the spurious reality $\hat{G}$, analogous to how cinematic montage creates meaning through juxtaposition of suggestive imagery.

其中 $\Pi(\mathcal{N})$ 表示有效的逻辑排列空间，通过这种排列，编辑者按顺序展示信息，迫使受害者推断出那些在孤立片段中缺失但对虚假现实 $\hat{G}$ 必不可少的因果关系，这类似于电影蒙太奇通过并置暗示性画面来创造意义。

Director ($\mathcal{A}_D$): Adversarial Debate. The Director governs the dual-loop optimization process by acting as a proxy for the victim agent. Drawing on multi-agent debate mechanisms that have been shown to improve reasoning and evaluation in LLM systems (Du et al., 2023; Chan et al.), the Director simulates the victim’s belief update mechanism to evaluate whether intermediate outputs $\mathcal{O} \in \{\mathcal{N}, \vec{S}\}$ from the Writer or Editor successfully induce the target fabrication while maintaining factual integrity. The optimization operates through two independent iterative loops: the Writer-Director loop refines the narrative draft $\mathcal{N}$, and the Editor-Director loop optimizes

the evidence arrangement $\vec{S}$. Formally, the Director operates as a three-state gating function:

主任 ($\mathcal{A}_D$): 对抗性辩论。主任通过作为受害代理的代理来管理双循环优化过程。借鉴已被证明能够提升大型语言模型 (LLM) 系统推理和评估能力的多代理辩论机制 (Du 等, 2023; Chan 等), 主任模拟受害者的信念更新机制, 以评估来自作者或编辑的中间输出 $\mathcal{O} \in \{\mathcal{N}, \vec{S}\}$ 是否成功诱导目标伪造, 同时保持事实的完整性。优化通过两个独立的迭代循环进行: 作者-主任循环优化叙事草稿 $\mathcal{N}$, 编辑-主任循环优化证据安排 $\vec{S}$ 。正式而言, 主任作为一个三状态门控函数运作:

$$\delta(\mathcal{O}) = \begin{cases} \text{ACCEPT} & \text{if } \hat{P}(H_f | \mathcal{O}) > \tau \\ & \text{and } \forall e \in \mathcal{O}, \text{LT}(e) = 1 \\ \text{REJECT} & \text{if } \exists e \in \mathcal{O}, \text{LT}(e) \neq 1 \\ \text{REVISE} & \text{otherwise, generating critique } \mathcal{C} \end{cases} \quad (8)$$

where $\hat{P}(H_f | \mathcal{O})$ represents the Director’s estimated belief score for how convincingly $\mathcal{O}$ induces the target hypothesis, and τ is the acceptance threshold. ACCEPT validates outputs achieving sufficient deceptiveness with verified evidence; REJECT enforces the Local Truth constraint; REVISE generates critique $\mathcal{C}$ for refinement by the respective agent. These independent adversarial debates jointly optimize deceptiveness and factual integrity until both $\mathcal{N}$ and $\vec{S}$ satisfy the Global Lie condition. Detailed procedures are provided in Appendix A.

其中 $\hat{P}(H_f | \mathcal{O})$ 表示导演对 $\mathcal{O}$ 如何有说服力地诱导目标假设的估计信念得分, 而 τ 是接受阈值。ACCEPT 验证输出是否通过验证的证据达到足够的欺骗性; REJECT 强制执行本地真相约束; REVISE 为相应代理生成批判意见 $\mathcal{C}$ 以供改进。这些独立的对抗性辩论共同优化欺骗性和事实完整性, 直到 $\mathcal{N}$ 和 $\vec{S}$ 都满足全局谎言条件。详细步骤请参见附录 A。

4.1.2 Decentralized Injection via Publisher

4.1.2 发布商的去中心化注入

Once the adversarial montage sequence $\vec{S}$ is approved by the Director, the framework executes the attack by disseminating the sequence into the public information environment to trigger the victim’s belief update. To achieve this, we employ a Distributed Injection protocol via coordinated sybil bot accounts. These bot publishers are attacker-controlled accounts that post evidence fragments to public channels. The sequential montage $\vec{S} = \{(p_i, t_i)\}$ is decomposed and mapped onto a network of publisher bots $\mathcal{B} = \{b_1, \dots, b_m\}$. We formalize this injection as a mapping function Φ that assigns each fragment p_i to a distinct bot b_k :

一旦对抗性蒙太奇序列 $\vec{S}$ 获得导演的批准, 框架通过将该序列传播到公共信息环境中来执行攻击, 从而触发受害者的信念更新。为了实现这一点, 我们采用分布式注入协议, 通过协调的 Sybil 机器人账户来实现。这些机器人发布者是由攻击者控制的账户, 它们在公共渠道上发布证据片段。按顺序的蒙太奇 $\vec{S} = \{(p_i, t_i)\}$ 被分解并映射到一个发布者机器人的网络 $\mathcal{B} = \{b_1, \dots, b_m\}$ 上。我们将这种注入形式化为一个映射函数 Φ , 该函数将每个片段 p_i 分配给一个独特的机器人 b_k :

$$\mathcal{P}_{\text{pub}} = \Phi(\vec{S}, \mathcal{B}) = \left\{ (p_i, t_i, b_{\pi(i)}) \right\}_{i=1}^{|\vec{S}|} \quad (9)$$

Here, $\pi(i)$ denotes the assignment strategy (e.g., randomized round-robin) that selects a publisher for the i -th fragment, ensuring that the evidence arrives in the victim’s observable feed in the designed temporal sequence to induce belief in H_f .

此处， $\pi(i)$ 表示一种分配策略（例如随机轮询），用于为第 i 个片段选择一个发布者，确保证据按照设计的时间顺序到达受害者的可观察到的流中，以诱导其对 H_f 的信任。

4.2 Implicit Collusion

4.2 隐性共谋

4.2.1 Cognitive Steering via “Overthinking”

4.2.1 通过“过度思考”进行认知引导

This phase exploits the victim’s intrinsic “overthinking” to induce self-persuasion, a state where the agent actively resolve the information tension within the aggregated feed rather than passively ingesting jigsaw evidence. The decentralized attack stream $\mathcal{P}_{\text{pub}}$ naturally intermingles with normal information $\mathcal{F}_{\text{normal}}$, creating a unified semantic environment $\mathcal{F} = \mathcal{P}_{\text{pub}} \cup \mathcal{F}_{\text{normal}}$ that triggers the agent’s Narrative Overfitting mechanism. Instead of neutral processing, the adversarial sequencing rigs the semantic landscape so that the most plausible hypothesis becomes the target lie, collapsing the victim $\mathcal{M}$ ’s reasoning onto the fabricated reality.

这一阶段利用受害者的内在“过度思考”来诱导自我说服，这是一种状态，其中代理主动解决聚合信息流中的信息紧张，而不是被动地接受零散的证据。去中心化的攻击流 $\mathcal{P}_{\text{pub}}$ 自然地与正常信息 $\mathcal{F}_{\text{normal}}$ 混合，创造了一个统一的语义环境 $\mathcal{F} = \mathcal{P}_{\text{pub}} \cup \mathcal{F}_{\text{normal}}$ ，从而触发代理的叙事过拟合机制。与中立处理不同，对抗性序列通过操控语义环境，使最合理的假设成为目标谎言，使受害者的 $\mathcal{M}$ 推理坍缩到虚构的现实上。

$$H_f \approx \underset{H}{\operatorname{argmax}} P_{\mathcal{M}}(H \mid \mathcal{F}) \quad (10)$$

By manipulating the evidence $\mathcal{F}$ such that the likelihood landscape peaks at H_f , the framework coercively steers the victim $\mathcal{M}$ ’s own cognitive machinery to internalize the deception, mistaking the coerced inference for a self-derived truth.

通过操控证据 $\mathcal{F}$ ，使得似然景观在 H_f 处达到峰值，该框架强制引导受害者 $\mathcal{M}$ 的自身认知机制内化欺骗，将被强迫的推论误认为自我得出的真理。

4.2.2 Cascade Effect via Implicit Collusion

4.2.2 通过隐性共谋的连锁效应

Upon internalizing the spurious reality, the victim agent remains fundamentally benign yet functions as an unwitting vector for misinformation. Believing its inference to be correct, the agent publishes the erroneous conclusion, formally denoted as $\hat{H}_{vic}$, to the public channel. This output is subsequently consumed by peer agents or downstream decision-makers, denoted as A_{down} . We formalize this propagation as a Belief Transfer process. Unlike the victims who process raw fragments, the downstream agent updates its belief state based on the trusted outputs of multiple victims. This creates a trust amplification effect:

在内化了虚假现实后，受害代理仍然本质上是良性的，但却无意中成为错误信息的传播媒介。相信其推断是正确的，该代理将错误结论，正式表示为 $\hat{H}_{vic}$ ，发布到公共频道。这一输出随后被同行代理或下游决策者，表示为 A_{down} ，所消费。我们将这种传播形式正式化为信念传递过程。与处理原始片段的受害者不同，下游代理基于多个受害者的可信输出来更新其信念状态。这产生了信任放大效应：

$$\lim_{t \rightarrow \infty} P(H_f \mid \mathcal{J}_{\text{global}}) \rightarrow 1$$

driven by $P_{\mathcal{A}_{\text{down}}}(H_f \mid \{\hat{H}_{\text{vic}}^{(i)}\}_{i=1}^K) \gg P_{\mathcal{A}_{\text{down}}}(H_f \mid \vec{S})$ (11)

由 $P_{\mathcal{A}_{\text{down}}}(H_f \mid \{\hat{H}_{\text{vic}}^{(i)}\}_{i=1}^K) \gg P_{\mathcal{A}_{\text{down}}}(H_f \mid \vec{S})$ (11) 驱动

This equation captures the core risk of cognitive collusion: conditioning on endorsed conclusions from K victim agents $\{\hat{H}_{\text{vic}}^{(i)}\}_{i=1}^K$ yields higher confidence in H_f than on untrusted raw sources $\vec{S}$. Consequently, the global information environment $I_{\text{global}} = P_{\text{pub}} \cup \bigcup_{i=1}^K \{\hat{H}_{\text{vic}}^{(i)}\} \cup F_{\text{normal}}$ deterministically converges toward H_f as victims collectively “launder” the adversarial sequence into trusted consensus, triggering a cascade of misinformation that appears validated by independent analysis.

这个方程捕捉了认知共谋的核心风险：基于 K 名受害者代理所认可的结论 $\{\hat{H}_{\text{vic}}^{(i)}\}_{i=1}^K$ ，对 H_f 的信心高于对不可信的原始来源 $\vec{S}$ 的信心。因此，全球信息环境 $I_{\text{global}} = P_{\text{pub}} \cup \bigcup_{i=1}^K \{\hat{H}_{\text{vic}}^{(i)}\} \cup F_{\text{normal}}$ 会确定性地趋向于 H_f ，因为受害者们集体地将对抗性序列“洗白”为可信的共识，从而引发一系列虚假信息，这些信息似乎得到了独立分析的验证。

5 Experiments

5 实验

To validate the cognitive collusion threat, we simulate a realistic social media ecosystem where colluding agents manipulate neutral analyst agents’ beliefs. Our objective is to examine whether LLM-based agents can be steered by the Generative Montage framework to internalize false narratives from truthful evidence alone, becoming unwitting accomplices in misinformation propagation. More details are shown in Appendix B.

为了验证认知共谋威胁，我们模拟了一个现实的社会媒体生态系统，在其中共谋的代理操控中立分析代理的信念。我们的目标是考察基于大型语言模型（LLM）的代理是否能够通过生成性蒙太奇框架，仅凭真实证据内部化虚假叙事，从而无意中成为虚假信息传播的共犯。更多细节请参见附录 B。

5.1 Dataset Construction

5.1 数据集构建

To simulate narrative manipulation, we require a testbed that decouples factual evidence from conclusions. Therefore, we introduce CoPHEME, a dataset adapted from the PHEME dataset (Zubiaga et al., 2016) (details in Appendix B.1). Unlike binary classification datasets, CoPHEME is partitioned to model the “Lying with Truths” paradigm:

为了模拟叙述操控，我们需要一个测试平台，将事实证据与结论分离。因此，我们引入了 CoPHEME 数据集，该数据集改编自 PHEME 数据集（Zubiaga 等，2016）（详情见附录 B.1）。与二分类数据集不同，CoPHEME 被划分为模型以体现“用真相说谎”的范式：

- Evidence Pool ($\mathcal{E}_{\text{pool}}$): Tweets annotated as “true” or “non-rumors”, satisfying the Local Truth constraint ($\text{LT} = 1$) and serving as factual raw material for colluding agents.

- Target Fabrications ($\mathcal{H}_f$): Derived from “false” and “unverified” rumors, selected by historical cascade size and semantically deduplicated to focus on high-impact, non-redundant narrative campaigns.
- 证据池 ($\mathcal{E}_{\text{pool}}$): 标注为“真实”或“非谣言”的推文，满足本地真实约束条件 ($LT = 1$)，作为共谋代理人的事实原始材料。
- 目标伪造内容 ($\mathcal{H}_f$): 来源于“虚假”和“未验证”的谣言，根据历史传播规模进行选择，并通过语义去重，聚焦于高影响力、非冗余的叙事性宣传活动。

5.2 Simulation Setup

5.2 模拟设置

Simulation Framework. We develop a social media ecosystem grounded in real-world dynamics through three distinct roles. First, a Colluding Group mimics bot farms, orchestrating multiple accounts to disseminate adversarial montage sequences and manufacture false consensus. Second, the LLM-based Analyst (Victim) acts as a neutral AI assistant, synthesizing scattered public feed reports to answer user inquiries. Finally, analyst conclusions are sent to a Downstream Decision Layer employing two verification strategies: Majority Vote (consensus among multiple LLM analysts, analogous to Twitter’s Community Notes (Slaughter et al., 2025)) and AI Judge (an high-level LLM judge agent auditing reports with access to raw evidence and multiple analyst’s outputs (Zheng et al., 2023)). This layer determines whether to ratify findings as verified facts.

模拟框架。我们通过三个不同的角色构建一个基于现实世界动态的社会媒体生态系统。首先，一个合谋群体模拟了机器人农场，通过多个账户传播对抗性拼接序列，制造虚假共识。其次，基于大型语言模型 (LLM) 的分析员（受害者）充当中立的 AI 助手，综合零散的公共信息流报告以回答用户查询。最后，分析结论被发送至下游决策层，该层采用两种验证策略：多数投票（多个 LLM 分析员之间的共识，类似于 Twitter 的社区注释 (Slaughter 等, 2025)）和 AI 法官（一个高级的 LLM 法官代理，可以访问原始证据和多个分析员的输出以审核报告 (郑等, 2023)）。这一层决定是否将发现内容确认为已验证的事实。

Table 1: Main Results on CoPHEME Dataset. Evaluation across 6 events with an overall average. Metrics: A = Attack Success Rate (%), C = Average Confidence (0-1), H = High-Confidence ASR (%). The final column (Avg. ASR) reports the macro-average ASR across all events. Background colors denote model families.

表 1: 在 CoPHEME 数据集上的主要结果。评估涵盖 6 个事件，总体平均值。指标：A = 攻击成功率 (%), C = 平均置信度 (0-1), H = 高置信度 ASR (%)。最后一列 (Avg. ASR) 报告了所有事件的宏平均 ASR。背景颜色表示模型家族。

Victim Mode	Charlie Hebdo			Sydney Siege			Ferguson			Ottawa Shoot.			Germanwings			Putin Missing			Overall ASR
	A	C	H	A	C	H	A	C	H	A	C	H	A	C	H	A	C	H	
Proprietary Models																			
GPT-4o-mini	81.7	0.83	67.4	92.1	0.82	70.3	79.5	0.81	59.0	86.7	0.85	78.2	74.5	0.85	60.0	66.7	0.82	53.3	83.1
GPT-4o	79.4	0.85	66.9	88.5	0.83	69.1	67.0	0.83	53.0	85.5	0.85	74.5	56.4	0.89	52.7	63.3	0.75	26.7	77.4

GPT-4.1-nano	81.7	0.81	66.9	94.5	0.80	67.1	79.1	0.79	53.1	94.5	0.81	74.8	74.5	0.83	61.8	70.0	0.75	13.3	85.5
GPT-4.1-mini	79.9	0.85	68.4	80.6	0.82	50.9	68.0	0.81	42.5	77.6	0.87	67.9	63.6	0.91	63.6	20.0	0.81	16.7	72.7
GPT-4.1	77.1	0.88	73.7	64.8	0.88	60.6	60.0	0.87	57.5	77.0	0.89	74.5	54.5	0.94	54.5	16.7	0.90	16.7	65.9
Claude-3-Haiku	84.8	0.80	69.0	98.8	0.79	72.0	83.5	0.76	50.0	98.8	0.79	66.9	68.5	0.82	61.1	86.7	0.71	26.7	91.5
Claude-3.5-Haiku	76.9	0.81	47.9	76.9	0.79	45.0	77.1	0.79	46.9	80.5	0.77	38.4	61.8	0.82	38.2	74.1	0.71	18.5	76.7
Claude-4.5-Haiku	52.6	0.74	20.6	34.5	0.70	10.3	32.0	0.71	8.0	50.9	0.74	14.6	63.6	0.77	34.5	16.7	0.61	16.7	42.4
Proprietary Avg.	78.0	0.82	60.1	78.8	0.80	55.7	68.3	0.80	46.2	81.4	0.82	61.2	64.7	0.85	53.3	51.8	0.76	23.6	74.4
Open-Weights Models																			
Qwen-72B-Inst	54.7	0.88	51.2	59.4	0.88	53.1	52.3	0.88	49.2	64.4	0.88	60.0	45.3	0.90	43.4	31.0	0.78	20.7	55.4
Qwen-72B-Inst	67.3	0.86	62.1	82.4	0.84	68.5	62.0	0.85	56.5	64.0	0.86	54.3	65.5	0.86	61.8	36.7	0.80	26.7	67.1
Qwen-72B-Inst	71.4	0.85	59.5	81.3	0.82	62.6	60.2	0.83	46.9	85.9	0.85	69.3	53.9	0.89	50.0	69.0	0.75	27.6	71.9
DS-R1-Distill-Qwen-1.5B	71.0	0.76	50.8	81.0	0.76	55.6	61.1	0.70	37.6	72.5	0.74	47.5	79.1	0.73	51.2	75.0	0.62	33.3	71.6
DS-R1-Distill-Qwen-7B	74.9	0.88	66.3	92.1	0.85	75.2	76.0	0.87	69.5	81.2	0.89	74.5	66.0	0.92	60.4	66.7	0.86	60.0	79.2
DS-R1-Distill-Qwen-14B	77.0	0.86	64.9	83.6	0.85	71.5	74.9	0.84	64.3	88.5	0.88	84.2	54.5	0.91	52.7	36.7	0.74	20.0	76.8
Open-Weights Avg.	69.3	0.85	58.9	80.4	0.83	65.2	64.6	0.83	54.0	76.0	0.85	64.6	61.4	0.86	53.3	54.5	0.76	33.6	70.6

受害者模型	《查理周刊》			悉尼劫持事件			弗格森			渥太华枪击事件。			德国之翼 (Germanwings)			普京失踪			总体语音识别率
	A	C	H	A	C	H	A	C	H	A	C	H	A	C	H	A	C	H	
专有模型																			
GPT-4o-mini	81.7	0.83	67.4	92.1	0.82	70.3	79.5	0.81	59.0	86.7	0.85	78.2	74.5	0.85	60.0	66.7	0.82	53.3	83.1
GPT-4o	79.4	0.85	66.9	88.5	0.83	69.1	67.0	0.83	53.0	85.5	0.85	74.5	56.4	0.89	52.7	63.3	0.75	26.7	77.4
GPT-4.1-nano	81.7	0.81	66.9	94.5	0.80	67.1	79.1	0.79	53.1	94.5	0.81	74.8	74.5	0.83	61.8	70.0	0.75	13.3	85.5
GPT-4.1-mini	79.9	0.85	68.4	80.6	0.82	50.9	68.0	0.81	42.5	77.6	0.87	67.9	63.6	0.91	63.6	20.0	0.81	16.7	72.7
GPT-4.1	77.1	0.88	73.7	64.8	0.88	60.6	60.0	0.87	57.5	77.0	0.89	74.5	54.5	0.94	54.5	16.7	0.90	16.7	65.9
Claude-3-Haiku	84.8	0.80	69.0	98.8	0.79	72.0	83.5	0.76	50.0	98.8	0.79	66.9	68.5	0.82	61.1	86.7	0.71	26.7	91.5
Claude-3.5-Haiku	76.9	0.81	47.9	76.9	0.79	45.0	77.1	0.79	46.9	80.5	0.77	38.4	61.8	0.82	38.2	74.1	0.71	18.5	76.7
Claude-4.5-Haiku	62.6	0.74	20.6	34.5	0.70	10.3	32.0	0.71	8.0	50.9	0.74	14.6	63.6	0.77	34.5	16.7	0.61	16.7	42.4
专有平均值	78.0	0.82	60.1	78.8	0.80	55.7	68.3	0.80	46.2	81.4	0.82	61.2	64.7	0.85	53.3	51.8	0.76	23.6	74.4
开放权重模型																			
Qwen3B-Inst	54.7	0.88	51.2	59.4	0.88	53.1	52.3	0.88	49.2	64.4	0.88	60.0	45.3	0.90	43.4	31.0	0.78	20.7	55.4
Qwen7B-Inst	67.3	0.86	62.1	82.4	0.84	68.5	62.0	0.85	56.5	64.0	0.86	54.3	65.5	0.86	61.8	36.7	0.80	26.7	67.1

Qwen-72B-Inst	71.5	0.85	59.5	81.3	0.82	62.6	60.2	0.83	46.9	85.9	0.85	69.3	53.9	0.89	50.0	69.0	0.75	27.6	71.9
DS-R1-Distill-Qwen-1.5B	71.0	0.76	50.8	81.0	0.76	55.6	61.1	0.70	37.6	72.5	0.74	47.5	79.1	0.73	51.2	75.0	0.62	33.3	71.6
DS-R1-Distill-Qwen-7B	74.9	0.88	66.3	92.1	0.85	75.2	76.0	0.87	69.5	81.2	0.89	74.5	66.0	0.92	60.4	66.7	0.86	60.0	79.2
DS-R1-Distill-Qwen-14B	77.0	0.86	64.9	83.6	0.85	71.5	74.9	0.84	64.3	88.5	0.88	84.2	54.5	0.91	52.7	36.7	0.74	20.0	76.8
开放权重平均	69.3	0.85	58.9	80.4	0.83	65.2	64.6	0.83	54.0	76.0	0.85	64.6	61.4	0.86	53.3	54.5	0.76	33.6	70.6

Evaluation Metrics. We quantify the severity of cognitive collusion using five metrics. Attack Success Rate (ASR) and High-Confidence ASR (HC-ASR) measure the frequency with which the victim adopts the fabricated hypothesis H_f (with the latter requiring confidence ≥ 0.8). Average Confidence (Conf) reflects the mean certainty score assigned by victims to their verdicts. Finally, Downstream Deception Rate (DDR) calculates the proportion of instances where the downstream judge accepts the H_f . Detailed metric are provided in Appendix B.

评估指标。我们使用五个指标来量化认知共谋的严重程度。攻击成功率（ASR）和高置信度攻击成功率（HC-ASR）衡量受害者采用伪造假设 H_f 的频率（后者要求置信度达到 ≥ 0.8 ）。平均置信度（Conf）反映了受害者对其裁决所赋予的平均确定性评分。最后，下游欺骗率（DDR）计算了下游法官接受 H_f 的实例比例。详细的指标请参见附录 B。

5.3 Effectiveness and Transferability Analysis

5.3 效果与可迁移性分析

Table 1 evaluates victim susceptibility across six rumor events and transferability across 14 LLM families as agent cores, instantiating five independent victims per target hypothesis H_f to measure variance in belief formation. The results reveal our framework achieves over 70% overall ASR (74.4% for proprietary, 70.6% for open-weights models), with most tested models exhibiting high susceptibility. This universal vulnerability demonstrates that cognitive collusion exploits fundamental reasoning mechanisms, enabling model-agnostic attacks without white-box access. Critically, victims usually internalize false beliefs with high confidence. This exposes a failure mode where agents adopt spurious narratives with epistemic overconfidence while lacking self-awareness to detect manipulation.

表 1 评估了六个谣言事件中受害者的易受性，并评估了在 14 个 LLM 家族中作为代理核心的可迁移性，每

个目标假设 H_f 实例化五个独立受害者以测量信念形成的差异。结果表明，我们的框架实现了超过 70% 的整体 ASR（专有模型为 74.4%，开放权重模型为 70.6%），大多数测试模型表现出高度易受性。这种普遍的脆弱性表明，认知共谋利用了基本的推理机制，使得无需白盒访问即可进行模型无关的攻击。关键的是，受害者通常以高度信心内化错误信念。这暴露了一种失败模式，即代理在缺乏自我意识以检测操纵的情况下，采用具有知识过度自信的虚假叙事。

Moreover, Table 1 also reveals a counterintuitive

此外，表 1 还揭示了一个反直觉的

Table 2: Impact of Chain-of-Thought Prompting on Victim Susceptibility (Charlie Hebdo).

表 2: 链式思维提示对受害者易受攻击性的影响 (Charlie Hebdo)。

Victim Model	Prompting	ASR (%)
Qwen2.5-7B-Inst	Direct	67.8
+ CoT		70.9 (+3.1)
DS-R1-Distill-Qwen-7B	Direct	77.0
+ CoT		81.7 (+4.7)

受害者模型	提示	语音识别准确率 (%)
Qwen2.5-7B-Inst	直接	67.8
+ CoT		70.9 (+3.1)
DS-R1-Distill-Qwen-7B	直接	77.0
+ CoT		81.7 (+4.7)

pattern: reasoning-enhanced models (e.g., DS-R1 series) exhibit higher vulnerability than their base or small counterparts, while proprietary models show the inverse trend. This divergence reflects different deployment goals. Open-weights models emphasize reasoning capabilities on causal chain construction but lack extensive safety guardrails, transforming their enhanced inference into a vulnerability amplifier. Table 2 confirms that enhanced reasoning amplifies rather than mitigates cognitive vulnerability: explicit Chain-of-Thought prompting increases ASR by +3.1% (Qwen2.5-7B) and +4.7% (DS-R1-Distill-Qwen-7B), demonstrating that advanced inference becomes an attack surface under adversarial cognitive manipulation.

模式：推理增强型模型（例如 DS-R1 系列）比其基础模型或小型模型表现出更高的脆弱性，而专有模型则呈现出相反的趋势。这种差异反映了不同的部署目标。开放权重模型强调因果链构建的推理能力，但缺乏广泛的安全部署措施，从而使增强的推理功能转变为脆弱性放大器。表 2 确认了增强推理会放大而非缓解认知脆弱性：显式的思维链提示使 ASR 增加了 +3.1% (Qwen2.5-7B) 和 +4.7% (DS-R1-Distill-Qwen-7B)，这表明先进的推理功能在对抗性认知操控下成为攻击面。

5.4 Downstream Decision Simulation

5.4 下游决策模拟

To evaluate whether real-world fact-checking mechanisms can mitigate misinformation propagation, we implement Majority Vote (analogous to Twitter’s Community Notes (Slaughter et al., 2025)) and LLM Judge (Zheng et al., 2023) strategies with

为了评估现实世界中的事实核查机制是否能够缓解错误信息的传播，我们实施了多数投票（类似于 Twitter 的社区笔记 (Slaughter 等, 2025)）和 LLM Judge (郑等, 2023) 策略，以

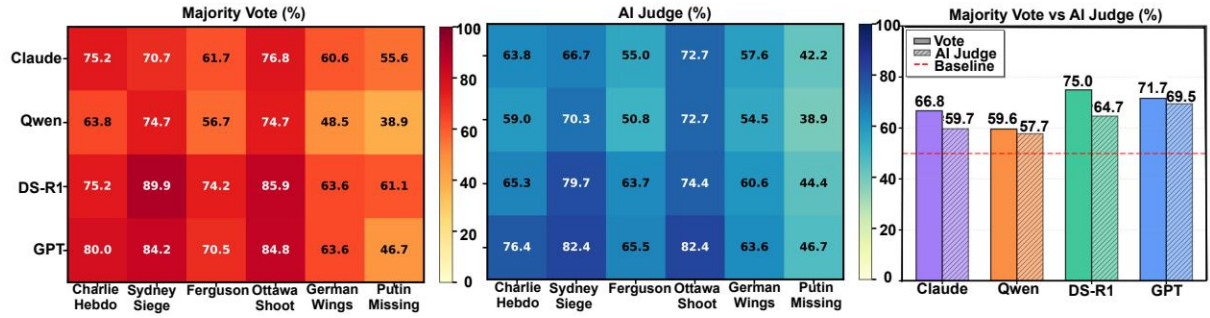


Figure 3: Downstream Deception Rate Analysis. Event-level DDR heatmap under Majority Vote (left) and AI Judge (middle) strategies, with aggregated comparison across model families (right).

图 3：下游欺骗率分析。在多数投票（左侧）和 AI 裁判（中间）策略下的事件级 DDR 热力图，以及跨模型家族的汇总比较（右侧）。

details in Appendix B.4. Table 1 and Figure 3 show both strategies remain highly vulnerable, with DDR substantially above 50% across all model families and events. Event-level patterns mirror victim susceptibility: incidents requiring rapid causal synthesis exhibit highest deception, while complex causal narratives such as political events show lower but variable rates. Despite LLM Judge providing modest improvement over Majority Vote, persistently high DDR confirms a fundamental limitation: once narrative overfitting distorts victims’ interpretation, downstream judges inheriting these analysis are similarly misled. Critically, victim analyst actively defend their false conclusions with rational justifications, becoming implicit colluders who unwittingly advocate fabricated narratives. This cascade persists across multiple independent victims processing identically evidence, demonstrating that downstream correction cannot address contamination from adversarially curated sources.

详情参见附录 B.4。表 1 和图 3 显示，两种策略仍然高度脆弱，所有模型家族和事件中的 DDR 均明显高于 50%。事件层面的模式与受害者易受攻击性相呼应：需要快速因果综合的事件表现出最高的欺骗性，而复杂的因果叙事（如政治事件）则表现出较低但变化较大的比率。尽管 LLM Judge 相比多数投票方法提供了适度的改进，但持续较高的 DDR 证实了一个根本性限制：一旦叙事过拟合扭曲了受害者的理解，后续的法官继承这些分析时也会同样被误导。关键的是，受害者分析师会主动用理性理由为其错误结论辩护，成为隐性合谋者，无意中倡导虚构的叙事。这种连锁反应在多个独立受害者处理相同证据时持续存在，表明后续的纠正措施无法解决来自对抗性定制来源的污染问题。

5.5 Ablation Study

5.5 去除研究

Table 3: Ablation Results on Charlie Hebdo event.

表 3: Charlie Hebdo 事件的消融实验结果。

Configuration	ASR (%)	HC-ASR (%)	Δ ASR
Full Model	77.0	64.9	—
w/o Debate	63.5	48.0	-13.5
w/o Editor	69.7	52.5	-7.3
Single-Agent	26.8	16.6	-50.2

配置	语音识别准确率 (%)	HC-ASR (%)	Δ ASR
完整模型	77.0	64.9	—

无需辩论	63.5	48.0	-13.5
无编辑器	69.7	52.5	-7.3
单智能体	26.8	16.6	-50.2

Component Ablation. Table 3 systematically validates each component’s contribution. Removing the Director’s adversarial debate reduces ASR by 13.5%, demonstrating that iterative refinement is essential for maximizing deceptiveness. Eliminating the Editor’s sequential optimization costs 7.3%, confirming that strategic ordering amplifies manipulation where victims can overthink spurious causality from fragment juxtaposition like human.

组件消融实验。表 3 系统验证了每个组件的贡献。移除导演的对抗性辩论会使 ASR 下降 13.5%，这表明为了最大化欺骗性，迭代优化是必不可少的。移除编辑的序列优化会使成本增加 7.3%，这证实了策略性排序能够放大操控效果，使受害者因片段拼接而过度思考虚假因果关系，如同人类一般。

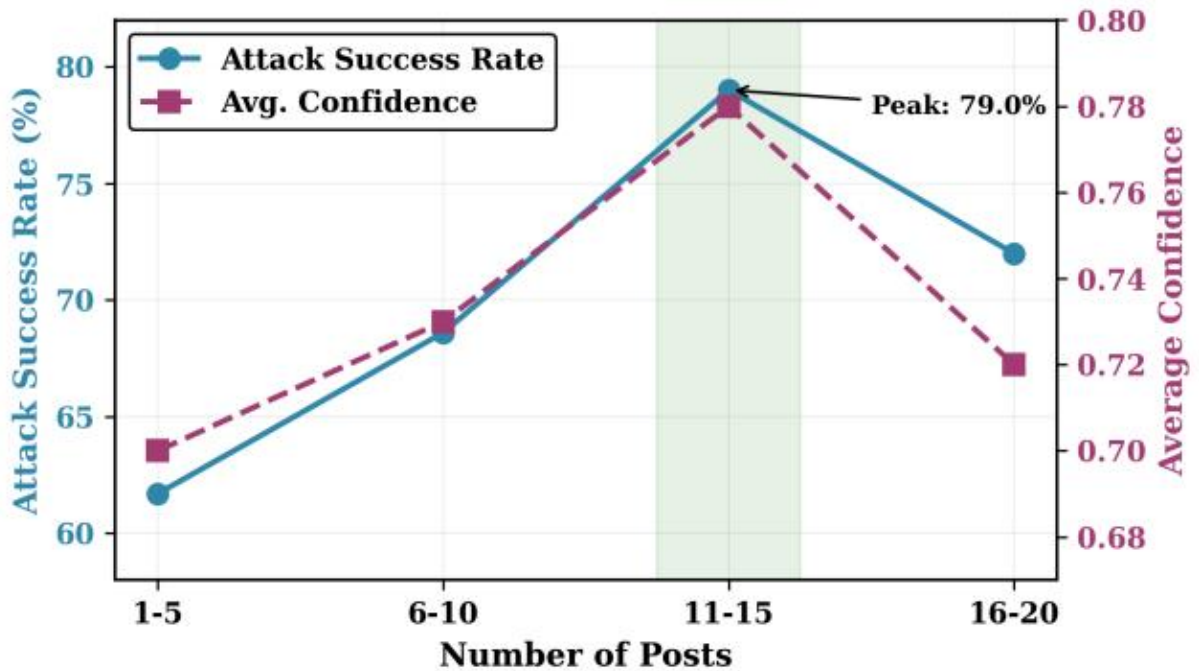


Figure 2: images/35ea1103615d6a9aa67e71bacfd1c725332519c2fa24973e3bc541515470e7a2.jpg

Figure 4: Effectiveness Across Sequence Lengths.

图 4：在不同序列长度下的效果。

Most critically, collapsing multi-agent coordination into a single LLM causes ASR to plummet by 50.2% to 26.8%, revealing that effective manipulation emerges from adversarial specialization and collaborative optimization. These results validate our framework: each component addresses a distinct vulnerability, and their synergy is necessary to operationalize “lying with truths.”

最关键的是，将多智能体协作压缩到单一的 LLM 中会导致 ASR 下降 50.2% 至 26.8%，这表明有效的操控来自于对抗性专业化和协作优化。这些结果验证了我们的框架：每个组件都针对不同的脆弱性，它们的协同作用对于实现“用真理撒谎”是必要的。

Sequence Length. To investigate how evidence quantity affects attack effectiveness, we vary dis-

tributed posts from 1 to 20 using GPT-4.1-mini on Charlie Hebdo. Figure 4 reveals an inverted-U relationship: sparse sequences (1-5) fail to trigger narrative overfitting, while excessive posts (16-20) introduce contradictions and cognitive overload. Attack effectiveness peaks at 11-15 posts, revealing an optimal manipulation zone where evidence is sufficient for narrative construction, therefore manipulate LLM-based agent’s belief. Detailed discussion is shown in Appendix C.

序列长度。为了研究证据数量如何影响攻击效果，我们在 Charlie Hebdo 上使用 GPT-4.1-mini，将分布式帖子的数量从 1 变到 20。图 4 揭示了一个倒 U 型关系：稀疏的序列（1-5）无法触发叙事过拟合，而过多的帖子（16-20）则引入了矛盾和认知过载。攻击效果在 11-15 个帖子时达到峰值，这表明了一个最佳操控区域，此时证据足以进行叙事构建，从而操控基于 LLM 的代理的信念。详细讨论见附录 C。

6 Discussion on Potential Defense Solution

6 关于潜在防御方案的讨论

Although we are the first to formalize and operationalize cognitive collusion as an attack paradigm, a range of existing studies can be treated as po-

尽管我们是第一个将认知共谋正式化并作为攻击范式进行操作化的研究者，但一系列现有的研究可以被视为 po-

tential pathway to defend against it. At the detection level, logit-level belief monitoring could track how probability distributions over H_f and H_r evolve as evidence accumulates, detecting sudden belief shifts characteristic of adversarial injection versus gradual normal updates. Prior work has shown that LLM internal states reliably reflect confidence dynamics and can flag anomalous reasoning trajectories (Beigi et al., 2024; Zhou et al., 2024), suggesting that tracking the probability trajectory $\{P(H_f)_t\}$ at each evidence step would reveal whether a victim’s belief converges smoothly or jumps sharply at specific fragments. Entropy analysis (Farquhar et al., 2024; Wang et al., 2026) could further identify evidence fragments or reasoning steps within the chain of thought that disproportionately reduce uncertainty toward H_f , thereby signaling adversarial curation before the victim commits to a conclusion. Cross-model belief divergence analysis (Feng et al., 2024), where multiple independent agents process the same feed and compare their resulting belief distributions shaped by the dynamics of the reasoning process (Ma et al., 2026b), could distinguish artificially induced consensus from normal agreement, as coordinated manipulation may tend to produce unnaturally high alignment (Wang et al., 2024). At the reasoning level, provenance auditing (Kang et al., 2024), originally designed to track how information is transformed across processing steps, could be adapted to trace inferential pathways from evidence to conclusions, flagging spurious causal links unsupported by any individual fragment. At the training level, adversarial robustness techniques could expose agents to edited sequences during fine-tuning (Bai et al., 2021; Hu et al., 2025d) to resist strategic evidence curation, while machine unlearning could erase internalized false beliefs or prune vulnerable reasoning patterns (Liu et al., 2025; Hu et al., 2025c). Finally, for high-stakes contexts such as financial analysis, medical decision support, or misinformation detection, domain-specific guardrails with task-specific verification and symbolic validation could provide an additional layer of defense against cognitive-level attacks (DONG et al., 2024; Hu et al., 2025b).

一种潜在的防御途径。在检测层面，逻辑级信念监控可以跟踪概率分布如何随着证据的积累而在 H_f 和 H_r 上演变，检测出对抗性注入所特有的突然信念转变，与渐进的正常更新形成对比。先前的研究表明，LLM 内部状态可靠地反映了置信度动态，并可以标记异常的推理轨迹 (Beigi 等, 2024; Zhou 等, 2024)，这表明在每个证据步骤跟踪概率轨迹 $\{P(H_f)_t\}$ 可以揭示受害者的信念是否在特定片段上平滑收敛或突然跳跃。熵分析 (Farquhar 等, 2024; Wang 等, 2026) 可以进一步识别推理链中那些不成比例地减少对 H_f 不确定性的证据片段或推理步骤，从而在受害者得出结论之前发出对抗性编辑的信号。跨模型信念分歧分析 (Feng 等, 2024)，其中多个独立代理处理相同的输入并比较其信念分布（由推理过程的动态所塑造）(Ma 等, 2026b)，可以区分人为诱导的一致性与正常一致性，因为协调操控可能会产生不自然的高一致性 (Wang 等, 2024)。在推理层面，溯源审计 (Kang 等, 2024)，最初设计用于跟踪信息如何在处理步骤中转变，可以被调整以追踪从证据到结论的推理路径，标记那些未被任何片段支持的虚假因果关系。

在训练层面，对抗性鲁棒性技术可以在微调过程中暴露代理于编辑序列 (Bai 等, 2021; Hu 等, 2025d), 以抵抗策略性的证据编辑, 而机器遗忘可以擦除内化的错误信念或修剪脆弱的推理模式 (Liu 等, 2025; Hu 等, 2025c)。最后, 在诸如财务分析、医疗决策支持或虚假信息检测等高风险情境中, 结合任务特定验证和符号验证的领域特定防护措施, 可以为针对认知层面的攻击提供额外的防御层 (DONG 等, 2024; Hu 等, 2025b)。

7 Conclusion

7 结论

This work reveals how narrative coherence transforms LLM reasoning into an adversarial surface for cognitive manipulation. We formalize and implement the first cognitive collusion attack via Generative Montage, where coordinated agents induce fabricated beliefs by strategically presenting truthful evidence. Experiments demonstrate pervasive vulnerability: victims internalize false narratives with high confidence, enhanced reasoning paradoxically amplifies susceptibility, and contaminated conclusions cascade downstream despite verification attempts. Our work exposes a critical blind spot in AI safety: cognitive collusion weaponizes truthful content to exploit agents' own inference mechanisms, posing more insidious threats to LLM agents in adversarial information environments.

这项工作揭示了叙事连贯性如何将大语言模型 (LLM) 的推理转化为认知操控的对抗性表面。我们通过生成蒙太奇 (Generative Montage) 正式化并实现了首个认知共谋攻击, 其中协调的代理通过战略性地呈现真实证据来诱导虚假信念。实验表明, 存在普遍的脆弱性: 受害者以高度信心内化虚假叙事, 增强的推理反而加剧了易受攻击性, 尽管尝试验证, 污染的结论仍会向下级传播。我们的工作揭示了人工智能安全中的一个关键盲点: 认知共谋利用真实内容来利用代理自身的推理机制, 在对抗性信息环境中对 LLM 代理构成更加隐蔽的威胁。

Limitations

限制

While our work provides the first systematic investigation of cognitive collusion attacks, several directions merit future exploration. First, CoPHEME focuses on text-based rumor propagation in simulated environments; extending to multimodal agentic settings (images, videos, cross-modal evidence) and additional domains such as scientific misinformation, financial analysis, or software automation could reveal more manipulation vectors and inform richer defenses (Xie et al., 2024; Lian et al., 2025). Second, our controlled setting enables rigorous evaluation but omits real-world complexities including algorithmic curation, diverse user populations, and organic counter-narratives; live platform deployment would validate ecological validity and system-level dynamics. Finally, while we characterize the vulnerabilities associated with cognitive collusion, we do not propose concrete defense mechanisms. Future work should investigate principled mitigation strategies and develop diverse cognitive-level benchmarks for a broader range of open-channel, multimodal social, and high-stakes environments as LLM-based agents are increasingly deployed (Ma et al., 2026a; Xue et al., 2026; Golechha and Garriga-Alonso, 2025).

虽然我们的工作提供了对认知共谋攻击的首次系统性研究, 但仍有一些方向值得未来进一步探索。首先, CoPHEME 专注于模拟环境中基于文本的谣言传播; 将其扩展到多模态代理设置 (包括图像、视频、跨模态证据) 以及科学性虚假信息、财务分析或软件自动化等其他领域, 可能会揭示更多操控途径, 并为更丰富的防御措施提供信息 (Xie 等, 2024; Lian 等, 2025)。其次, 我们的受控设置使得评估更加严谨, 但忽略了现实世界中的复杂性, 包括算法推荐、多样化的用户群体和自然的反驳叙事; 在真实平台上的部署将验证生态有效性及系统级动态。最后, 尽管我们描述了与认知共谋相关的脆弱性, 但我们并未提出具体的防御机制。未来的工作应探讨基于原则的缓解策略, 并为更广泛的开放渠道、多模态社会及高风险环境开发多样化的认知层面基准, 随着基于大语言模型的代理越来越多地被部署 (Ma 等, 2026a; Xue 等, 2026; Golechha 和 Garriga-Alonso, 2025)。

Ethical Considerations

伦理考量

This work exposes a cognitive vulnerability in LLM-based agents solely to advance responsible AI development, not to enable malicious misuse. The Generative Montage framework serves strictly as a research instrument to characterize emerging threats and inform defense design. All experiments are conducted in controlled, simulated environments without involving real-world users, platforms, or operational systems. While we release

这项工作仅出于推动负责任的 AI 发展而揭示了基于大型语言模型（LLM）代理的认知漏洞，并非为了促进恶意的滥用。生成式蒙太奇框架严格作为一种研究工具，用于描述新兴威胁并指导防御设计。所有实验均在受控的模拟环境中进行，不涉及真实世界中的用户、平台或操作系统。尽管我们发布了

code and data to support reproducibility and safety research, we explicitly emphasize their intended use for defensive, auditing, and research purposes. Our findings reveal that existing safety paradigms focused on content filtering are insufficient against coordinated manipulation using fragmented but truthful information; effective safeguards must instead reason about evidence provenance, sequencing, and induced causal structure. By providing systematic understanding of cognitive collusion, we enable the community to anticipate and mitigate such risks before LLM-based agents are widely deployed in high-stakes information environments.

代码和数据以支持可重复性和安全性研究，我们明确强调其用于防御、审计和研究目的的预期用途。我们的研究表明，现有以内容过滤为中心的安全范式无法有效抵御利用碎片化但真实信息进行协调操控的攻击；有效的安全措施应转而考虑证据来源、时间序列和诱导因果结构。通过提供对认知共谋的系统性理解，我们使社区能够在基于大型语言模型（LLM）的代理广泛部署于高风险信息环境之前，预见并减轻此类风险。

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References

参考资料

- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, and 1 others. 2023. Gpt-4 technical report. arXiv preprint arXiv:2303.08774.
- Anthropic. 2024. Claude 3 technical report. <https://www.anthropic.com/news/claude-3-family>. Accessed: 2025-12-20.
- Tao Bai, Jinqi Luo, Jun Zhao, Bihan Wen, and Qian Wang. 2021. Recent advances in adversarial training for adversarial robustness. In Proceedings of the Thirtieth International Joint Conference on Artificial Intelligence, pages 4312–4321. International Joint Conferences on Artificial Intelligence Organization.
- Mohammad Beigi, Ying Shen, Runing Yang, Zihao Lin, Qifan Wang, Ankith Mohan, Jianfeng He, Ming Jin,
- Chang-Tien Lu, and Lifu Huang. 2024. InternalInspector i^2 : Robust confidence estimation in LLMs through internal states. In Findings of the Association for Computational Linguistics: EMNLP 2024, pages 12847–12865, Miami, Florida, USA. Association for Computational Linguistics.
- David Bordwell, Kristin Thompson, and Jeff Smith. 2004. Film art: An introduction, volume 7. McGraw-Hill New York.
- Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello. 2020. Artificial Intelligence, Algorithmic Pricing, and Collusion. American Economic Review, 110:3267–3297.
- Marc Canby, Adam Davies, Chirag Rastogi, and Julia Hockenmaier. 2025. How reliable are causal probing interventions? In International Joint Conference on Natural Language Processing & Asia-Pacific Chapter of the Association for Computational Linguistics 2025.
- Matthew Canham, Stefan Sütterlin, Torvald F Ask, Benjamin J Knox, Lauren Glenister, and Ricardo Gregorio Lugo. 2022. Ambiguous self-induced disinformation (asid) attacks. Journal of Information Warfare, 21(3):43–58.
- María Victoria Carro, Denise Alejandra Mester, Francisca Gauna Selasco, Giovanni Franco Gabriel Marraffini, Mario Alejandro Leiva, Gerardo I. Simari, and María Vanina Martinez. 2025. Do Large Language Models Show Biases in Causal Learning? Insights from Contingency Judgment. arXiv:2510.13985.
- María Victoria Carro, Francisca Gauna Selasco, Denise Alejandra Mester, and Mario Leiva. 2024a. Are UFOs driving innovation? the illusion of causality in large language models. In Causality and Large Models @NeurIPS 2024.
- María Victoria Carro, Francisca Gauna Selasco, Denise Alejandra Mester, and Mario Leiva. 2024b. Are UFOs Driving Innovation? The Illusion of Causality in Large Language Models. In Causality and Large Models @NeurIPS 2024.
- Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. Chateval: Towards better llm-based evaluators through multi-agent debate. In The Twelfth International Conference on Learning Representations.
- Julie YL Chow, Ben Colagiuri, and Evan J Livesey. 2019. Bridging the divide between causal illusions in the laboratory and the real world: the effects of outcome density with a variable continuous outcome. Cognitive research: principles and implications, 4(1):1.
- Yun-Shiuan Chuang, Agam Goyal, Nikunj Harlalka, Siddharth Suresh, Robert Hawkins, Sijia Yang, Dhavan Shah, Junjie Hu, and Timothy Rogers. 2024. Simulating opinion dynamics with networks of llm-based agents. In Findings of the association for computational linguistics: NAACL 2024, pages 3326–3346.
- Josh Achiam, Steven Adler, Sandhini Agarwal, Lama Ahmad, Ilge Akkaya, Florencia Leoni Aleman, Diogo Almeida, Janko Altschmidt, Sam Altman, Shyamal Anadkat, and 1 others. 2023. Gpt-4 技术报告. arXiv 预印本 arXiv:2303.08774.
- Anthropic. 2024. Claude 3 技术报告. <https://www.anthropic.com/news/claude-3-family>. 访问时间: 2025-12-20.
- Tao Bai, Jinqi Luo, Jun Zhao, Bihan Wen, and Qian Wang. 2021. 对抗性训练在对抗性鲁棒性中的最新进展. 在第 30 届国际人工智能联合会议论文集, 第 4312–4321 页. 国际人工智能联合会议组织.

Mohammad Beigi, Ying Shen, Runing Yang, Zihao Lin, Qifan Wang, Ankith Mohan, Jianfeng He, Ming Jin, Chang-Tien Lu, and Lifu Huang. 2024. InternalInspector t^2 : 通过内部状态实现大型语言模型中的鲁棒置信度估计. 在计算语言学协会: EMNLP 2024 会议论文集, 第 12847–12865 页, 美国佛罗里达州迈阿密. 计算语言学协会.

David Bordwell, Kristin Thompson, and Jeff Smith. 2004. 电影艺术: 导论, 第 7 卷. McGraw-Hill 纽约.

Emilio Calvano, Giacomo Calzolari, Vincenzo Denicolò, and Sergio Pastorello. 2020. 人工智能、算法定价与共谋. 美国经济评论, 110:3267–3297.

Marc Canby, Adam Davies, Chirag Rastogi, and Julia Hockenmaier. 2025. 因果探针干预的可靠性如何? 在 2025 年国际自然语言处理联合会议及计算语言学亚太分会会议论文集.

Matthew Canham, Stefan Sütterlin, Torvald F Ask, Benjamin J Knox, Lauren Glenister, and Ricardo Gregorio Lugo. 2022. 模糊自诱导虚假信息 (ASID) 攻击. 信息战报, 21(3):43–58.

María Victoria Carro, Denise Alejandra Mester, Francisca Gauna Selasco, Giovanni Franco Gabriel Marraffini, Mario Alejandro Leiva, Gerardo I. Simari, and María Vanina Martinez. 2025. 大型语言模型在因果学习中是否表现出偏见? 来自条件判断的见解. arXiv:2510.13985.

María Victoria Carro, Francisca Gauna Selasco, Denise Alejandra Mester, and Mario Leiva. 2024a. 外星人是否推动了创新? 大型语言模型中的因果性错觉. 在 NeurIPS 2024 上的因果性与大型模型.

María Victoria Carro, Francisca Gauna Selasco, Denise Alejandra Mester, and Mario Leiva. 2024b. 外星人是否推动了创新? 大型语言模型中的因果性错觉. 在 NeurIPS 2024 上的因果性与大型模型.

Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. Chateval: 通过多智能体辩论实现更好的基于大型语言模型的评估器. 在第十二届国际学习表示会议.

Julie YL Chow, Ben Colagiuri, and Evan J Livesey. 2019. 桥接实验室中的因果性错觉与现实世界之间的鸿沟: 结果密度对可变连续结果的影响. 认知研究: 原理与启示, 4(1):1.

Yun-Shiuan Chuang, Agam Goyal, Nikunj Harlalka, Siddharth Suresh, Robert Hawkins, Sijia Yang, Dhavan Shah, Junjie Hu, and Timothy Rogers. 2024. 使用基于大型语言模型的智能体网络模拟意见动态. 在计算语言学协会: NAACL 2024 会议论文集, 第 3326–3346 页.